



Car Workshop Management System

Course:

Software Engineering

Instructor:

Majed Mohammed Al Saeed

Team Members:

Ali hussain Albather 223044348

Kumail Abdulraouf Aljaber 223023721

Mugdad alhammad 223029809

Saud AlAbdulrahim 223012272

Table of Contents

1.Introduction

2.User Requirements Definition

2.1 Customer Requirements

2.2 Mechanic Requirements

2.3 Admin Requirements

3.System Architecture

4. Functional Requirements

4.1 Customer Requirements

4.2 Mechanic Requirements

4.3 Admin Requirements

5. Non-Functional Requirements

5.1 Performance Requirements

5.2 Security Requirements

5.3 Usability Requirements

5.4 Compliance Requirements

5.5 Reliability Requirements

6. Process Model

6.1 Use Case Diagram

7. Use Case Diagram

8. Context model

9. Process model

10. Interaction model “sequence diagram”

11. Structural model “class diagram”

12. State machine diagram

13. Test cases

14. Tasks, durations, and dependencies

15. Activity bar chart

16. Staff allocation bar chart

1.Introduction

This document outlines the requirements and architectural design for the Car Workshop Management System. Developed as a small-scale application, the system aims to simplify essential workshop processes such as service booking, job assignment, and report generation. It is structured using a three-layer architecture—comprising the interface layer, business-logic layer, and data-handling layer—to ensure clarity, maintainability, and efficiency.

The system allows customers to register, log in, book appointments, and view the status of those appointments along with mechanics’ comments. Customers can also manage their accounts and send feedback to the admin. Mechanics can view and update assigned jobs, and admins can manage customer records, appointments, and reports, as well as create invoices.

User Types and Functionalities

User type	Key Responsibilities
customer	- Register and log in -Book service appointment - View the status of ongoing services and access service history. - Manage account details.

mechanic	- View assigned jobs - Update job progress Add brief comments on service progress.
admin	- Manage customer records. - Manage appointments - Generate reports and invoices. - Assign jobs.

2. User Requirements Definition

- Customer Requirements

- 1. Registration and Login:**
Customers can create accounts, ensuring secure access to personal information.
- 2. Appointment Booking:**
The capability to book appointments tailored to their schedule.
- 3. Service Tracking:**
Users can view current service statuses and retrieve their service history.

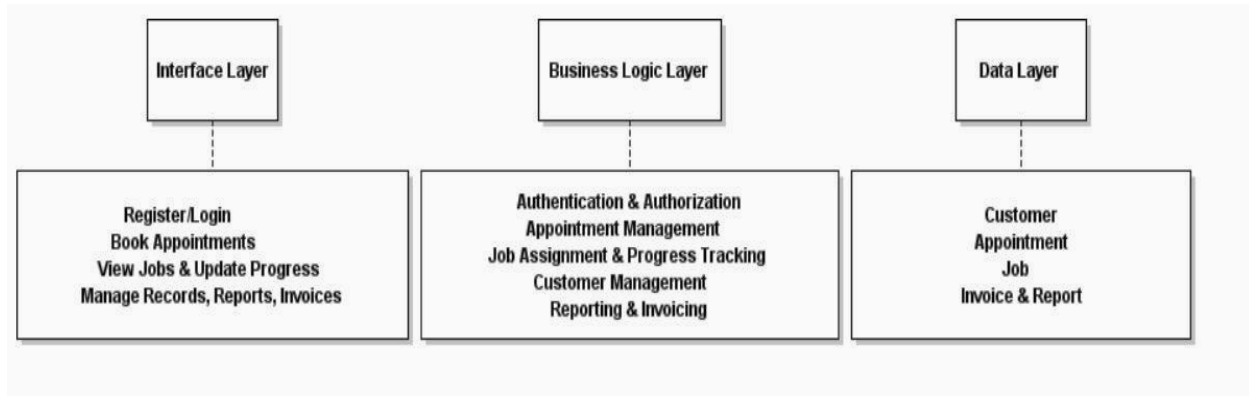
Mechanic Requirements

- **Job Assignment View:**
Mechanics should have access to a list of jobs assigned to them, facilitating better workload management.
- **Progress Updates:**
Mechanics should have the ability to mark jobs as "In Progress," "Completed," or "Pending" with additional comments as necessary.

Admin Requirements

- **Customer Management:**
Admins should be able to add, edit, or delete customer profiles and appointment entries and assign jobs and give access to mechanics
 - **Reporting and Invoicing:**
The system must support the generation of detailed reports and invoices to aid in business operations.
-

3. System Architecture



4. Functional Requirements

4.1 Customer Requirements

- **FR-1:** The system shall allow customers to register and log in
- **FR-2:** The system shall enable customers to book service appointments by selecting date and time.
- **FR-3:** The system shall allow customers to view the status of their ongoing services and review their service history.

4.2 Mechanic Requirements

- **FR-4:** The system shall allow mechanics to view a list of jobs assigned to them.
- **FR-5:** The system shall allow mechanics to update job progress by marking jobs as "Pending," "In Progress," or "Completed," and add brief comments when necessary.

4.3 Admin Requirements

- **FR-6:** The system shall allow admins to manage customer accounts, including the ability to add, edit, and delete profiles.
- **FR-7:** The system shall allow admins to manage appointments overall, including cancellations and rescheduling.
- **FR-8:** The system shall provide functionality to generate reports on various metrics, including service history and customer feedback.

- **FR-9:** The system shall allow admins to assign service tasks to mechanics and monitor their completion status.

5. Non-Functional Requirements

5.1 Performance Requirements

- **NFR-1:** The system should support at least 30 concurrent users without performance degradation.

5.2 Security Requirements

1. **NFR-2:** All user data must be encrypted during transmission and storage.
2. **NFR-3:** The system shall implement role-based access control to restrict data access to authorized users.

5.3 Usability Requirements

1. **NFR-4:** The system's cli must be intuitive and user-friendly, requiring minimal training for users.

5.4 Compliance Requirements

- **NFR-5:** The system shall comply with relevant data protection regulations and allow customer to delete their account with data

5.5 Reliability Requirements

- **NFR6:** The system shall ensure consistent performance during normal operations without unexpected crashes.

6. Process Model

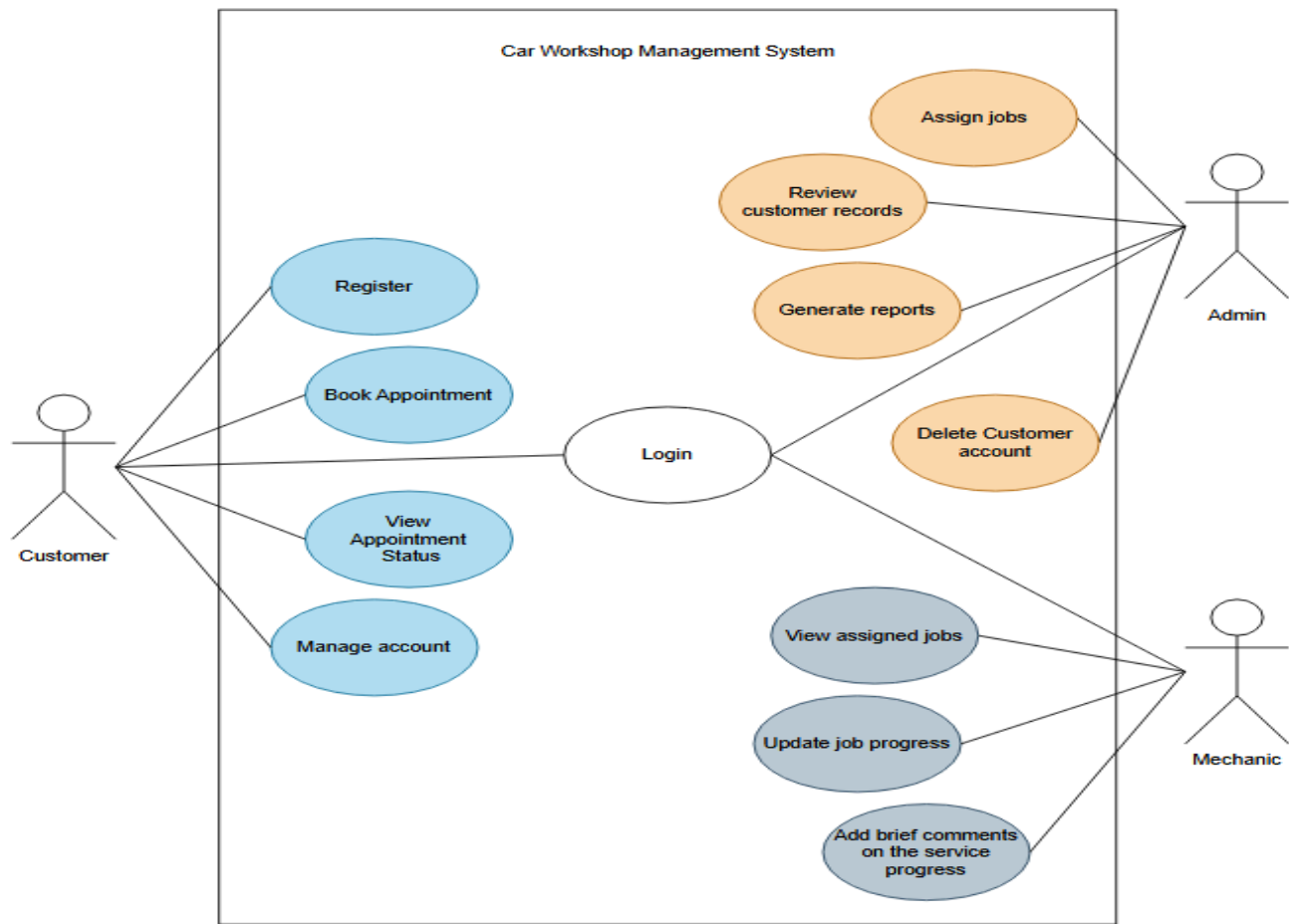
The **Waterfall process model** is chosen for the development of the **Car Workshop Management System**.

Although the Waterfall model is often used for **large-scale software projects** that require detailed documentation and well-defined stages, it is also suitable for this system because our requirements are **clear, stable, and unlikely to change** during development.

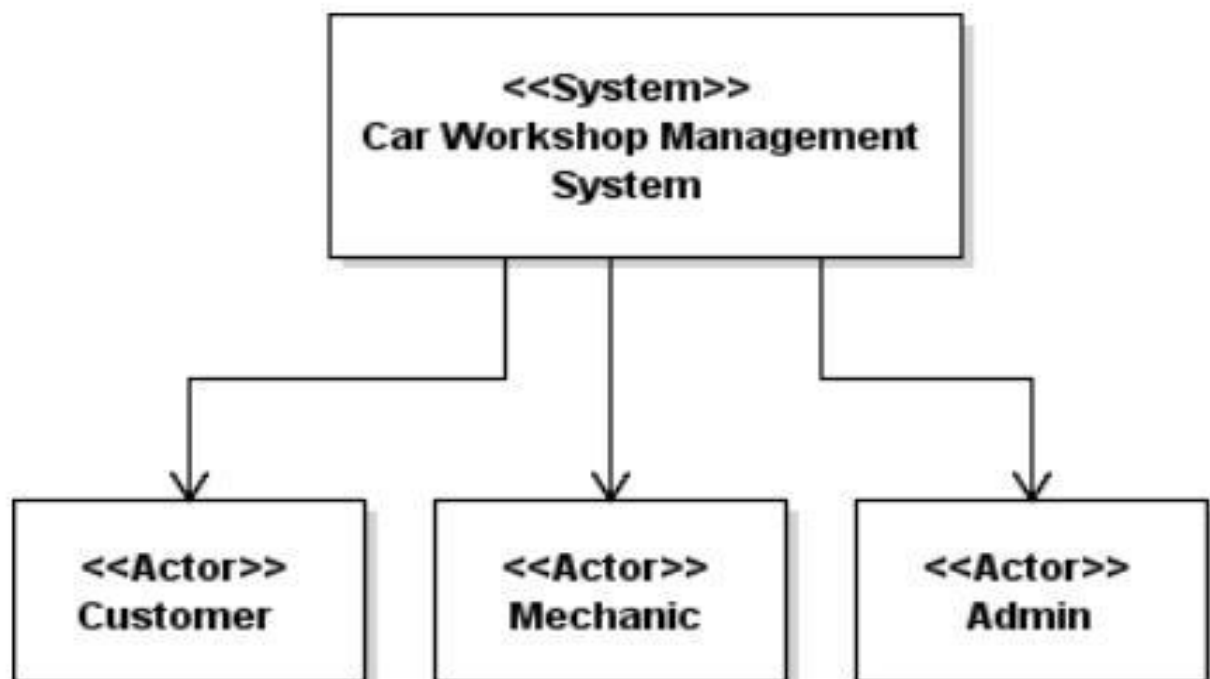
The project follows a small and well-understood scope, making it practical to apply a simplified version of the Waterfall approach. Each stage—**requirements analysis, system design, implementation, testing, and deployment**—is completed sequentially, ensuring that every phase is verified before moving on to the next.

By adopting this model, the team can maintain a **structured workflow**, produce organized documentation, and ensure that all requirements for the Car Workshop Management System are thoroughly implemented and tested before completion.

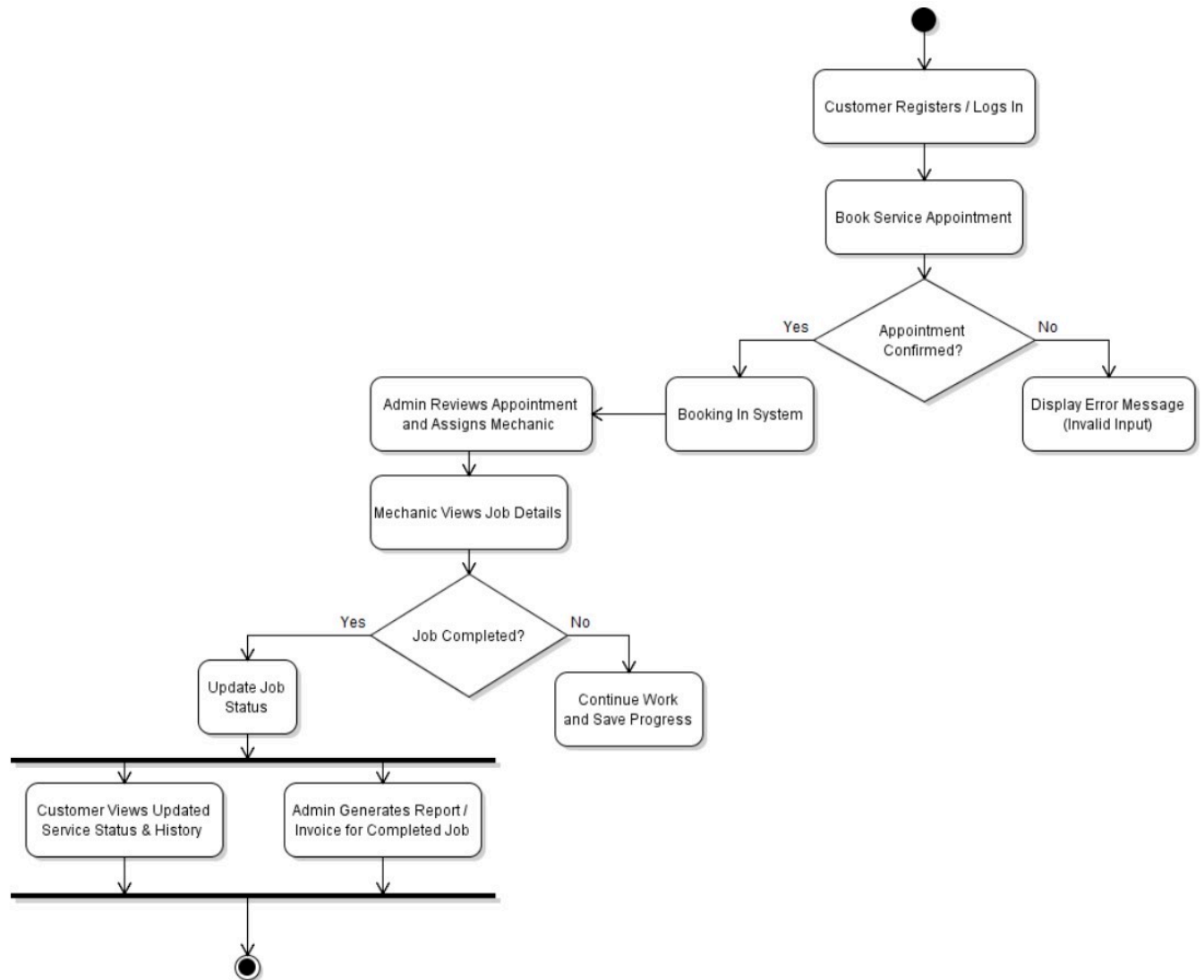
7.Use Case Diagram:



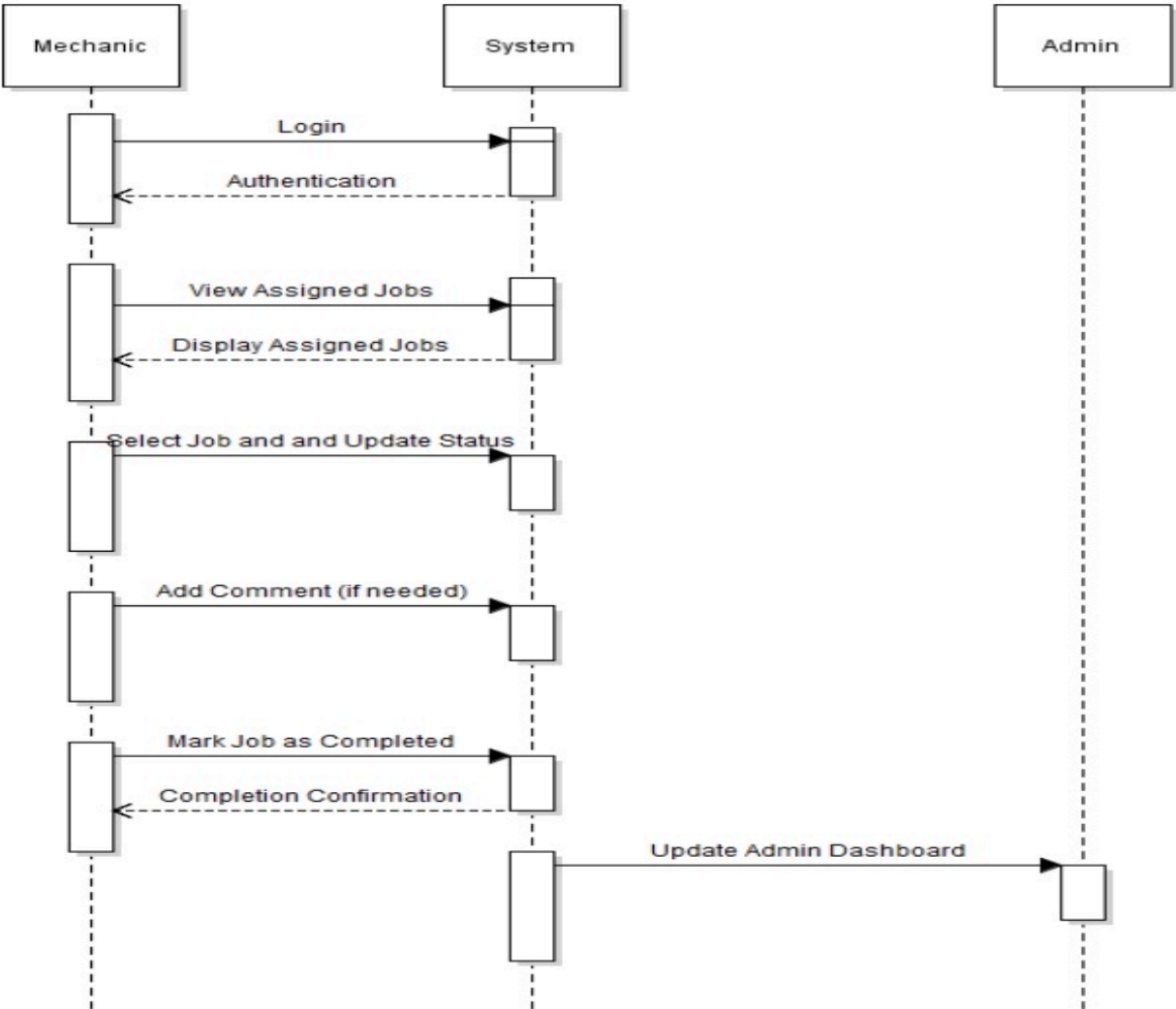
8.Context model

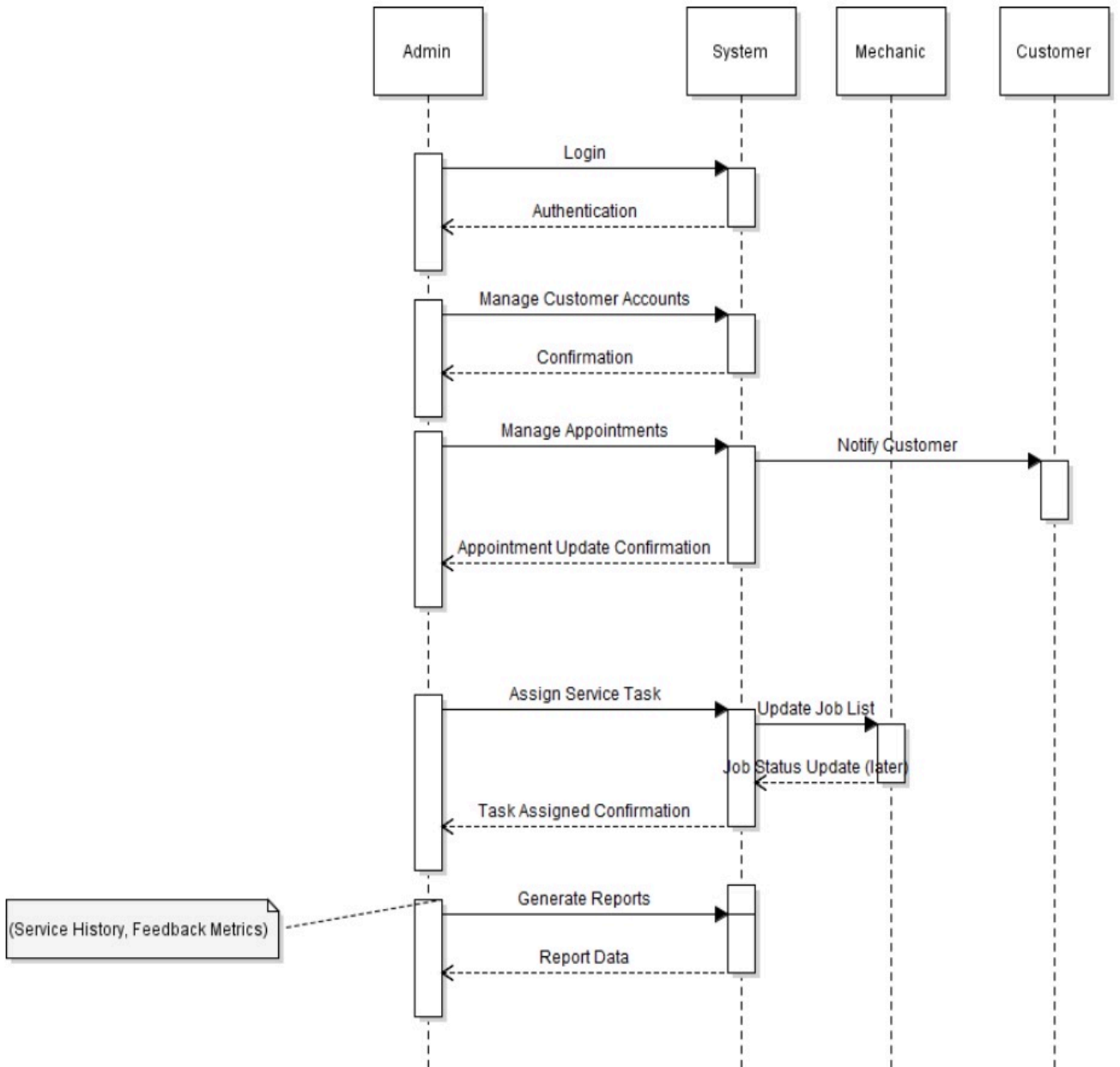


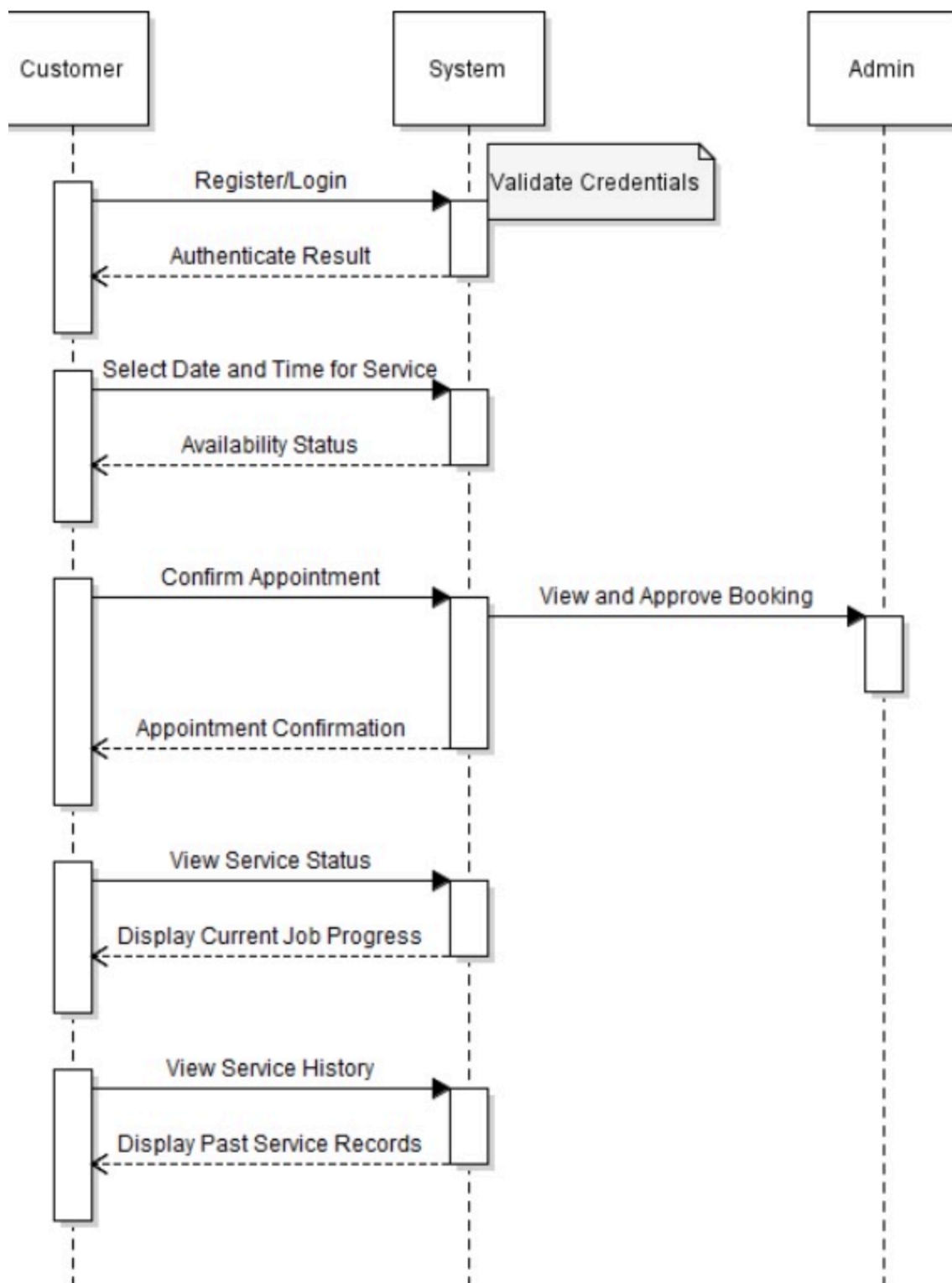
9.Process model



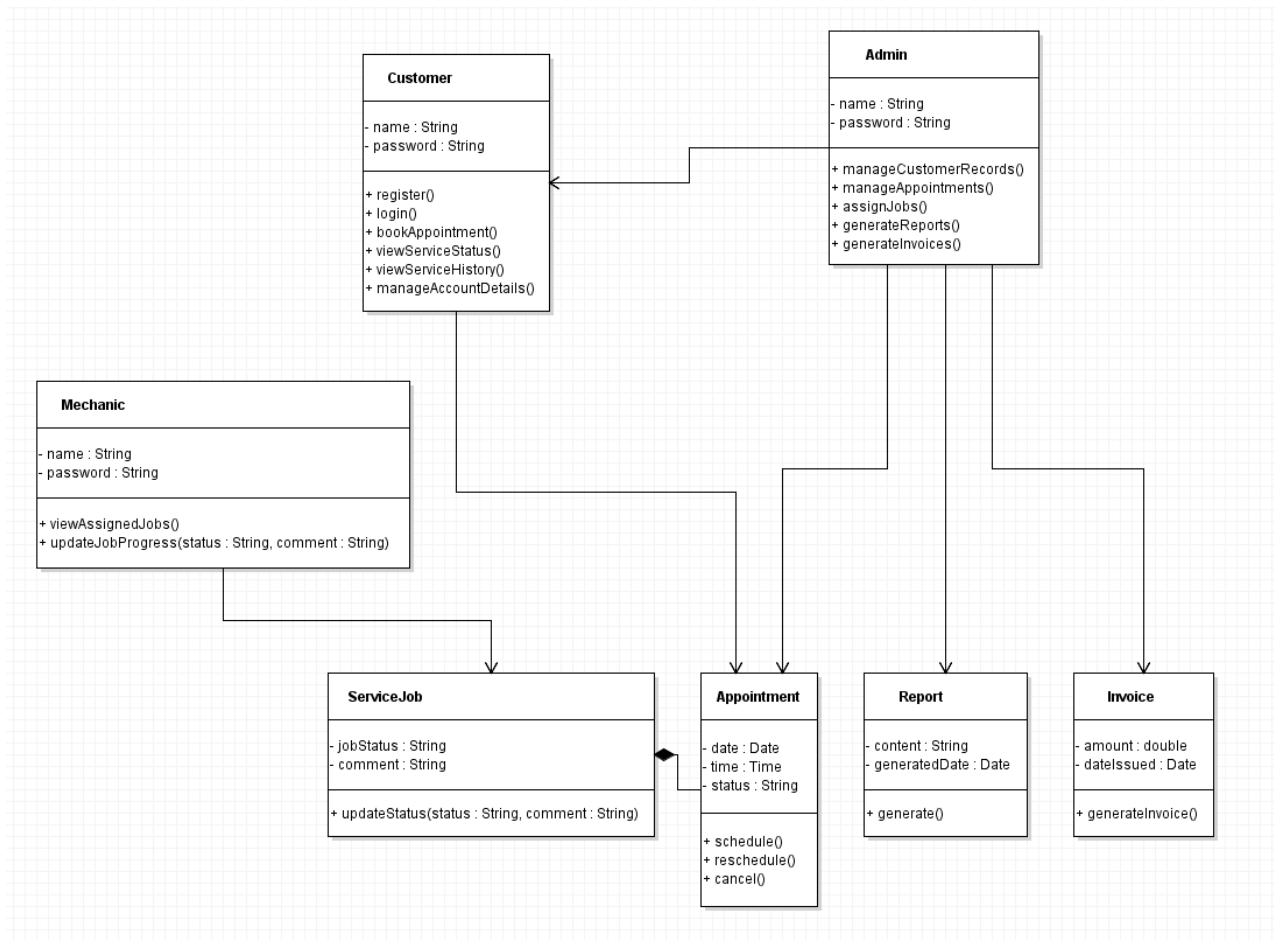
10 Interaction model “sequence diagram”







11.Structural model “class diagram”



12.State machine diagram



13. Test cases

Test Case: Customer-01 - Customer Registration, Logging in, Booking, and Tracking.

Objective: verify the entire customer workflow, ensuring a user can register, book a new service, and immediately confirm the service is correctly logged in their tracking history, including handling invalid booking attempts. (Validates FR-1, FR-2, FR-3).

Scenario:

1. **Registration (FR-1):** A new user registers successfully with the username "Customer1" and a complementary password.
2. **Valid Logging in and Booking (FR-2):** Customer1 logs in and attempts to book a service appointment for "2025-12-30" at "09:30". The system **shall** accept this booking and assign it a "Pending" status.
3. **Duplicate Rejection (FR-2):** Customer immediately attempts to book another service for the exact same date and time slot ("2025-12-30" at "09:30"). The system **shall** check and reject this attempt, displaying "You already have a booking at that date and time." and check only one job is created.
4. **Tracking (FR-3):** Customer1 navigates to his job list. The system **shall** display the single successful booking with the correct date, time, and the initial status of "Pending,"
Checking successful logging and tracking capabilities.

Test Case: Mechanic-01 - View only assigned jobs and update job status

Objective: To check that a Mechanic can correctly view only their assigned jobs (FR-4) and execute a status change, including adding service comments (FR-5).

Preconditions:

3. Customer "Customer1" has a job (Job ID 1) assigned to Mechanic "mech1." (Status: In Progress).
4. Customer "Customer2" has a job (Job ID 2) assigned to Mechanic "mech2." (Status: In progress).
5. Mechanic "mech1" is logged into the system.

Scenario:

1. **View Assigned Jobs (FR-4):** Mechanic "mech1" navigates to the 'View My Jobs' menu. Check that the system **shall only** display Job ID 1 (Customer1's job) and **shall not** display Job ID 2 (Customer2's job). Essential details like Customer Name, Date, Time and Status ("In Progress") **shall** be visible.
2. **Update Job Status (FR-5):** mech1 selects Job ID 1 for an update.
3. mech1 enters "**Completed**" as the new status.
4. mech1 enters the required service comment: "Replaced oil filter and changed oil"
5. **Verification:** Check that the system **shall** successfully update Job ID 1 to the "Completed" status and store the exact comment. Check when mech1 views the jobs again (FR-4), Job ID 1's summary **must** reflect the final "Completed" status.

Test Case: ADMN-01 - Admin Management, Assignment, and Reporting (FR-6, FR-7, FR-9, FR-8)

Objective: To check the Admin's capabilities to manage accounts, delegate work, and ensure reporting metrics are accurate after a sequence of operational changes.

Preconditions:

1. Customer "Customer1" exists with 1 pending jobs: Job 1 (Date X, Time Y).
2. Customer "Customer2" exists with 1 pending jobs: Job 2 (Date X, Time Y).
3. Customer "Customer3" exists with 1 pending jobs: Job 3 (Date Z, Time W).
4. Mechanics "mech1" and "mech2" are available.
5. Admin is logged into the system.

Scenario:

2. **Assignment (FR-9):** The Admin assigns Job 1 (Date X, Time Y) to "mech1" (Status moves to "In Progress"). **Check** when the Admin attempts to assign Job 2 (Date X, Time Y) to "mech1," the system **shall** detect a scheduling conflict and reject the assignment, forcing the Admin to choose "mech2" instead.
3. **Customer Management (FR-6/FR-7):** Check that the Admin view all customer records (FR-6) and confirms the assigned status of Job 1 and Job 2. The Admin then decides to delete a different, unrelated customer account, "Customer3" (FR-6/FR-7). **Check** that the system **shall** successfully delete "Customer3" and all jobs previously associated with "Customer3" **must** be removed from the global job list and any assigned mechanic's list (cascading delete).
4. **Reporting (FR-8):** **Check** that The Admin can generate the service report. The report **shall** accurately reflect the current state: total jobs reduced by the number of "Customer3" deleted jobs, a correct count for "In Progress" (Job 1 and Job 2), and a workload breakdown showing mech1 and mech2 each with one assigned job.

Test Case: End to End - Full Service Job Lifecycle Validation (FR-1, FR-2, FR-3, FR-4, FR-5, FR-9)

Objective: To verify and check the end-to-end process from a customer booking an appointment through the Admin's assignment and the Mechanic's completion, culminating in the customer viewing the final service history.

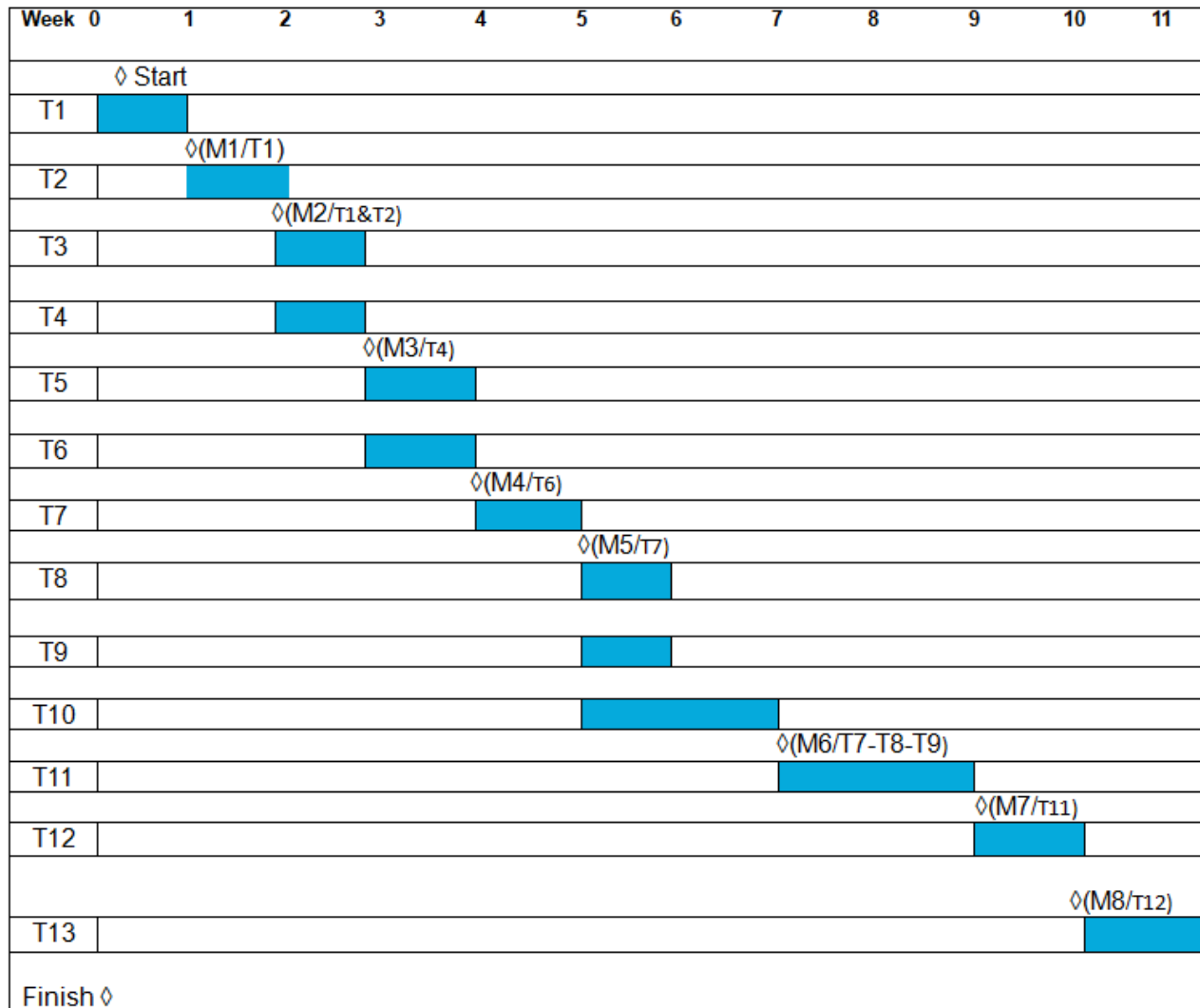
Scenario:

4. **Customer Action:** A new customer, "Customer1" registers (FR-1) and successfully books a service appointment for a future date (FR-2). The system records the job with a "Pending" status.
5. **Admin Action:** The Admin logs in, identifies Customer1's new job, and uses the assignment functionality (FR-9) to delegate the task to "mech1" which automatically transitions the job status to "In Progress."
6. **Mechanic Action:** mech1 logs in (FR-4) and views his assigned job list. He selects Customer1's job, changes the status to "**Completed**" (FR-5), and adds mandatory comments detailing the work performed (e.g., "Full oil change and tire change complete.").
7. **Verification:** Customer1 logs back in and checks his service tracking (FR-3). The system **shall** correctly display the job as "Completed" and indicate that the service was performed by "mech1" and include the full service comment, confirming successful data cascade across all three user roles.

14.Tasks, durations, and dependencies

Task	Duration (days)	Dependencies
Introduction(T1)	5	
User requirements definition(T2)	5	T1(M1)
System architecture(T3)	5	T1-T2(M2)
System requirements specification(T4)	5	T1-T2(M2)
Use case diagram (T5)	5	T4-(M3)
Context model (T6)	5	T4-(M3)
Process model(T7)	5	T6(M4)
Sequence diagrams(T8)	5	T7(M5)
Class diagram(T9)	5	T7(M5)
State machine diagram(T10)	10	T7(M5)
Implementation(T11)	15	T7-T8-T9(M6)
Test Cases(T12)	5	T11(M7)
Charts(T13)	5	T12(M8)

15. Activity bar chart



16.Staff allocation bar chart

